

Review

# Towards BIM-Based Sustainable Structural Design Optimization: A Systematic Review and Industry Perspective

Muhammad Afzal <sup>1,\*</sup> , Rita Yi Man Li <sup>2</sup> , Muhammad Faisal Ayyub <sup>3</sup>, Muhammad Shoaib <sup>1</sup> and Muhammad Bilal <sup>4</sup>

<sup>1</sup> Department of Architecture, Built Environment and Construction Engineering (DABCE), Politecnico di Milano, 20133 Milano, Italy; muhammad1.shoaib@mail.polimi.it

<sup>2</sup> Sustainable Real Estate Research Center, Department of Economics and Finance, Hong Kong Shue Yan University, North Point, Hong Kong 999077, China; ymli@hksyu.edu

<sup>3</sup> Department of Civil, Chemical, Environmental, and Material Engineering—DICAM, Alma Mater Studiorum, University of Bologna, 40126 Bologna, Italy; muhammad.faisal3@studio.unibo.it

<sup>4</sup> Department of Construction Engineering and Management, National University of Science and Technology (NUST), Islamabad 44000, Pakistan

\* Correspondence: muhammad.afzal@mail.polimi.it

**Abstract:** Structural design optimization (SDO) plays a pivotal role in enhancing various aspects of construction projects, including design quality, cost efficiency, safety, and structural reliability. Recent endeavors in academia and industry have sought to harness the potential of building information modeling (BIM) and optimization algorithms to optimize SDO and improve design outcomes. This review paper aims to synthesize these efforts, shedding light on how SDO contributes to project coordination. Furthermore, the integration of sustainability considerations and the application of innovative technologies and optimization algorithms in SDO necessitate more interactive early stage collaboration among project stakeholders. This study offers a comprehensive exploration of contemporary research in integrated SDO employing BIM and optimization algorithms. It commences with an exploratory investigation, employing both qualitative and quantitative analysis techniques following the PRISMA systematic review methodology. Subsequently, an open-ended opinion survey was conducted among construction industry professionals in Europe. This survey yields valuable insights into the coordination challenges and potential solutions arising from technological shifts and interoperability concerns associated with the widespread implementation of SDO. These preliminary steps of systematic review and industry survey furnish a robust foundation of knowledge, enabling the proposal of an intelligent framework for automating early stage sustainable structural design optimization (ESSDO) within the construction sector. The ESSDO framework addresses the challenges of fragmented collaboration between architects and structural engineers. This proposed framework seamlessly integrates with the BIM platform, i.e., Autodesk Revit for architects. It extracts crucial architectural data and transfers it to the structural design and analysis platform, i.e., Autodesk Robot Structural Analysis (RSA), for structural engineers via the visual programming tool Dynamo. Once the optimization occurs, optimal outcomes are visualized within BIM environments. This visualization elevates interactive collaborations between architects and engineers, facilitating automation throughout the workflow and smoother information exchange.

**Keywords:** automated structural design; building information modeling (BIM); design automation; generative design; interoperability; structural design optimization (SDO); systematic framework



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## 1. Introduction

Structural design represents a critical aspect of architecture, engineering, and construction (AEC) projects, entailing the creation of robust, secure, and enduring structures capable of withstanding diverse loads and environmental conditions [1]. The quality, effectiveness, performance, and environmental sustainability of construction projects, especially buildings

and infrastructure facilities, are usually influenced by the quality of the structural design of these structures and infrastructures. The global market for structural engineering reached a value of USD 10.4 billion in 2019 and was forecasted to attain USD 14.4 billion by 2027, displaying a compound annual growth rate (CAGR) of 4.1% from 2020 to 2027 [2]. Due to this enormous investment and the paramount significance of structural integrity within this realm, recent advancements in computational design and theoretical frameworks have advanced structural design practices [3]. This evolution has ushered in novel perspectives for the construction industry, ushering in automation and innovative design methodologies. The convergence of computer technology, modern structural engineering, and advances in construction materials has led to structural design optimization (SDO) [4] and structural design automation [5], with both enhancing the efficiency of AEC practices. Moreover, the utilization of these computational tools and methodologies in structural design has prompted AEC stakeholders to increasingly consider crucial environmental sustainability factors during the initial phases of building structure design [6]. These practices have urged the field of structural engineering to be advanced in terms of computational design and innovative frameworks. As such, these transformative shifts in structural engineering design have given rise to the automation and optimization of structural design and its strong association with several considerations that could be incorporated, including cost, integrity, durability, and sustainability.

### *1.1. Recent Advances in Structural Design Optimization (SDO)*

Recent developments in SDO have emphasized automating and optimizing structural configurations and dimensions to meet a variety of performance objectives. These advancements aim to optimize the configuration and dimensions of structures to optimize augmenting strength [7,8], minimize material usage [9,10], reduce costs [11], enhance energy efficiency [12,13], improve sustainability [14–17], and optimize several other performance criteria [12,18]. Concurrently, structural design automation endeavors to streamline the design process, mitigate human errors, and enhance productivity through computer-based tools and optimization algorithms. Prominent practices and technologies in this domain include the parametric design [7,19], generative design [20,21], building information modelling (BIM) technology [22–25], machine learning (ML), and artificial intelligence (AI) [21,26,27], as well as integrating finite element analysis (FEA) with simulation tools [28,29]. Parametric modeling techniques empower structural designers to define parameters governing a structure's geometry and dimensions, facilitating rapid exploration of diverse design variations and their performance analysis. Similarly, generative design employs algorithms to automatically generate designs based on predefined constraints and objectives. BIM, a transformative technology in AEC, facilitates 3D model creation and management, data sharing among stakeholders, and automates structural design [5]. AI algorithms and automated design and simulation tools enrich structural practitioners with intelligent platforms to analyze extensive data, predict structural behavior, and expedite design alternatives, ensuring alignment with envisaged structural performance.

These recent computational strides heighten the technological adoption during early design phases, and they have offered new prospects for the AEC industry [30]. According to a McKinsey & Company study, firms integrating digital technologies in engineering and construction could shorten project delivery times by 20% and achieve cost savings of up to 15% [31]. Beyond design automation, intricate structural designs necessitate digitally driven decision support tools that are stable, cost-effective, and aligned with project objectives. The synergy between design automation and optimization algorithms proves especially advantageous in the initial phases of structural design [30]. This synergy accelerates the design workflow while delivering optimal design solutions, recognizing the pivotal influence architects and structural designers wield in shaping a built asset's performance throughout its lifecycle. Consequently, there is a growing emphasis on early stage optimization to create optimal design solutions. Various studies have endeavored to automate structural design optimization during early stage conceptualization, enhancing

dependability, structural integrity, economic efficiency, and environmental sustainability of built environments. For instance, Mangal and Cheng [9] employed a hybrid genetic algorithm coupled with BIM technologies to automate detailed RC structure design. Eleftheriadis et al. [14] utilized BIM and the finite element method (FEM) to optimize flat slabs and reduce building material usage, which was mainly steel reinforcement. Another study by [4] integrated BIM and metaheuristic algorithms for RC structural design optimization, lowering steel reinforcement in high-rise structures and decreasing overall construction costs. Numerous studies [19,23,32–35] integrated various optimization strategies with BIM, presenting promising computational avenues for complex early stage structural engineering challenges. Looking more closely at the current state of SDO practices, it becomes clear that the role of BIM, generative design, and automation, technology, and artificial intelligence is transforming structural engineering procedures. As such, optimized structural designs enable reduced costs and enhance efficiency, material conservation, and sustainability enhancements, as well as allow designers and practitioners to explore many design variations quickly, resulting in high-quality and practical outcomes.

### *1.2. Challenges in the Current SDO Practices*

Digitally driven SDO practices offer numerous advantages to stakeholders, particularly architects and structural engineers, throughout the entire lifespan of a built asset. These practices leverage automation and optimization to unlock creative design possibilities for architects, while engineers focus on ensuring structural integrity and efficiency. However, the seamless integration between these two groups is a significant but often overlooked challenge. Advanced SDO practices are beneficial but can complicate the coordination between architects and structural engineers [36].

Moreover, while optimization, automation, and sustainability integration in SDO processes yield substantial benefits, they can introduce complexities in the coordination between architects and structural engineers. These complexities encompass interoperability issues stemming from the integration of diverse tools and platforms, intricate decision making that involves tradeoffs between structural performance and sustainability objectives, a learning curve as stakeholders adapt to new technologies, data-sharing challenges, and the need to align diverse objectives relating to aesthetics, structural integrity, and sustainability. Regrettably, existing research fails to provide a comprehensive and proactive approach to address these complexities, particularly in fostering collaboration among stakeholders in SDO processes. While some studies have touched upon related concerns [5,37–40], they primarily focused on integrating structural engineering considerations into architectural processes or vice versa, or they concentrated on collaboration during digital construction stages.

### *1.3. Research Significance*

Numerous research studies [23,30,41–47] have made significant efforts to address the aforementioned challenges prevalent in contemporary SDO practices. They have approached these issues from various angles and dimensions. Nonetheless, a noticeable research gap persists, specifically within the realm of BIM-based structural design optimization and automation during the initial stages of construction. This gap becomes particularly pronounced when considering the interactive collaboration between architects and structural engineers. In this context, Hamidavi et al. [23] has already highlighted the challenge of interoperability between these disciplines during automation and optimization. It is imperative to recognize that, while automation and optimization are vital, facilitating real-time collaborative interaction is equally indispensable. Automating design changes without allowing architects and engineers to iteratively adjust parameters and observe immediate structural modifications could hinder holistic, collaborative automation. To bridge this gap, it is crucial to enhance interoperability by automating platforms, fostering continuous communication, enabling joint exploration and decision making, and maximizing the potential for BIM-based design optimization and automation. Additionally,

SDO practices play a pivotal role in synergizing automation and optimization during the initial phases of structural design, underscoring the central role of architects and structural engineers in shaping the future of construction.

To address these research gaps, this study employs a systematic approach for a comprehensive review of the current State-of-the-Art concepts, offering both quantitative and qualitative analyses. This study rigorously evaluates existing research with a focus on early stage structural design optimization using emerging technologies. The quantitative analysis furnishes a statistical overview of pertinent research documents retrieved from scientific databases through specific keyword-based search queries. Concurrently, the qualitative analysis categorizes research documents into distinct objectives of structural design optimization tailored to different project phases and process levels across the building lifecycle. These categories encompass the *conceptualization and configuration stage*, *automated code compliance stage*, *fabrication and prefabrication details*, *construction execution stage*, and *structural monitoring phase*. To supplement this research investigation, an exploratory study includes an online survey of accredited structural engineers, primarily from Italy and Europe. This survey garners insights on the challenges of integration from an industry perspective. By integrating the results from quantitative and qualitative analyses along with the survey, our study proposes an intelligent framework for BIM-based early stage sustainable structural design optimization (ESSDO). This ESSDO framework seamlessly integrates BIM platforms for architecture and structural design using Autodesk Revit for architectural design and Autodesk Robot Structural Analysis (RSA) for structural design and analysis, using the visual programming tool Dynamo. The framework facilitates design, analysis, and optimization by generating optimal design configurations and promoting interactive collaboration between architects and engineers. This framework is adaptable to various structures, from tall buildings to residential units and bridges, and effectively addresses the challenges in the early stages of design.

#### 1.4. Research Questions and Objectives

This holistic literature review study mainly targets to answer the following research questions (RQs):

- (RQ<sub>1</sub>) What are the current research trends in automated structural design optimization (SDO) efforts?
- (RQ<sub>2</sub>) How do automation, optimization, and sustainability inclusion aspects during SDO affect the interactive coordination among architects and structural engineers for decision making?

To comprehensively answer these research questions, the following list comprises the research objectives/tasks (RTs) that will be executed in this research study.

- (RT<sub>1</sub>) Systematic analysis of the present state of research on the automation of structural design optimization (SDO).
- (RT<sub>2</sub>) Provision of quantitative and qualitative analyses of the current State-of-the-Art concepts in automating structural design optimization.
- (RT<sub>3</sub>) Exploration of challenges for collaboration and interoperability between architects and structural engineers for structural design optimization with the use of an online opinion survey.
- (RT<sub>4</sub>) Proposal of a systematic BIM-based framework for early stage sustainable structural design optimization (ESSDO) to streamline interactive coordination between architects and structural engineers.

#### 1.5. Paper Organization

- The overall structure of this study is organized as follows. Following this introduction, Section 2 consists of the “*methodology and literature retrieval*” (RT<sub>1</sub>) from the scientific databases. Research findings and discussions on the overview of the “*quantitative analysis of the current status*” (RT<sub>2</sub>) are then provided in Section 3. This section also consists of the “*qualitative analysis of the research and categorization of the available research*”

(RT<sub>2</sub>) based on the project phases and process levels. “*Opinion survey results from professionally acclaimed structural engineers*” (RT<sub>3</sub>) are also provided in Section 3. The “*ESSDO framework proposal*” (RT<sub>4</sub>) is presented in Section 4, followed by the “*research gaps and future scope*” highlighted in Section 5. Lastly, Section 6 “*concludes*” this research study along and outlines the “*limitations*” of this research for future studies.

## 2. Materials and Methods

This study embarks on an exhaustive retrieval and examination of research content concerning the automation of structural design optimization employing digital tools from 2010 to 2023. The initial objectives of this study, i.e., RT<sub>1</sub> and RT<sub>2</sub>, are directed toward furnishing quantitative and qualitative analyses of the current State-of-the-Art concepts. These analyses form the foundation for subsequent objectives, including an opinion survey and the development of the ESSDO framework. In light of this, a systematic literature review (SLR) is employed, as it adheres to a meticulous and specified procedure for identifying, assessing, and synthesizing the existing body of knowledge on a specific subject [48]. This approach serves to establish the current State-of-the-Art concepts in the field of discussion, i.e., the automation of structural design optimization through digital tools and optimization approaches. A detailed examination of the SLR process unveils a three-phase framework comprising planning, implementation, and reporting. The initial planning phase involves crafting pertinent research queries and setting specific criteria to facilitate the identification of relevant research articles and search strategies. The subsequent implementation stage encompasses gathering and selecting pertinent literature for incorporation into this study. Lastly, the reporting phase entails synthesizing and thoroughly analyzing the gathered literature. Therefore, in this research process, the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) method is adopted to collect the data for this systematic review. The widespread acceptance of this methodology technique in systematic review studies is attributed to its lucid delineation of the rationale and methodologies employed to locate, screen, exclude, and include the pertinent literature [49,50], bolstering the precision and accuracy of the systematic review process. It has been applied in the housing safety and health research [51], healthcare [52], and education [53].

### 2.1. Categorization and Scope Criteria

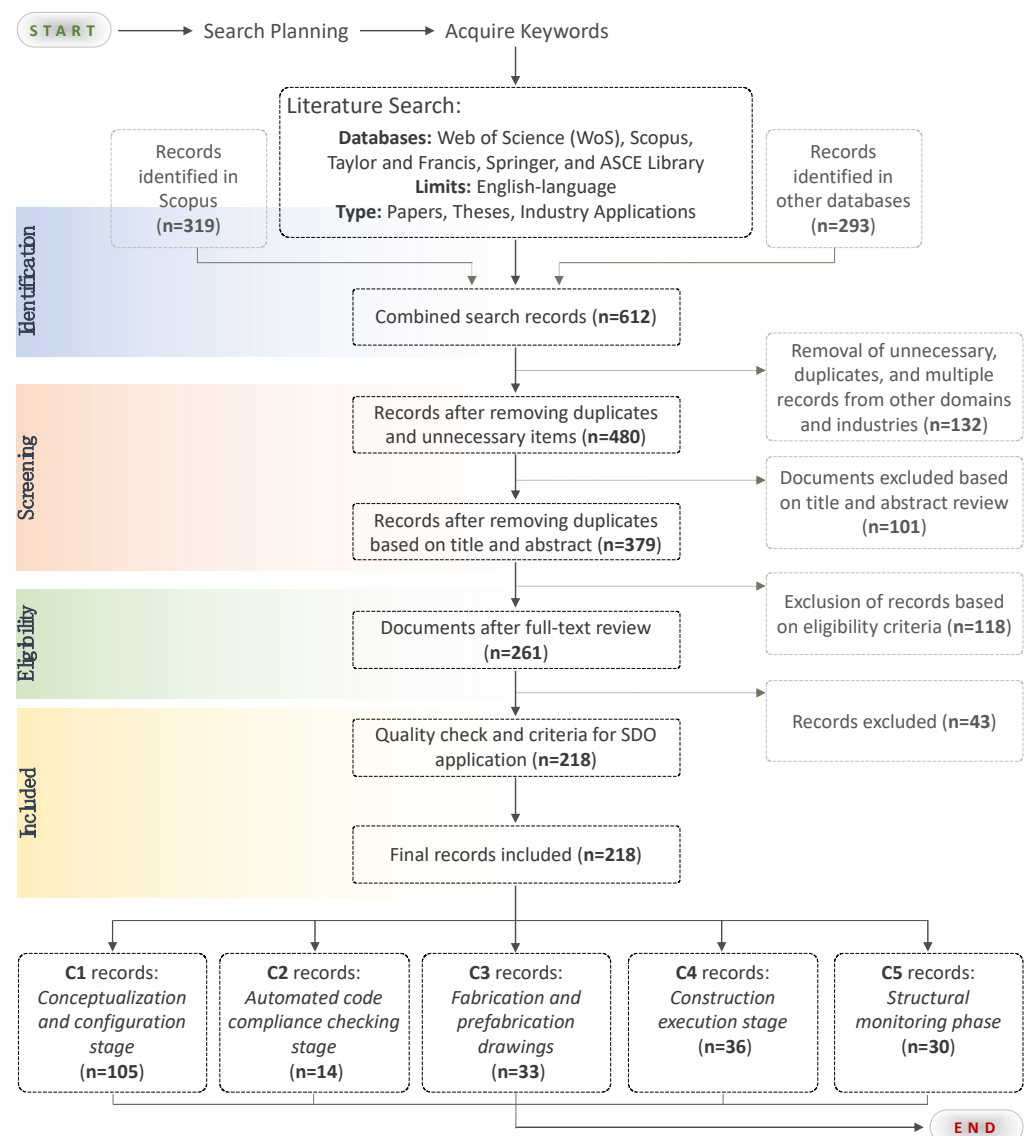
The principal categorization criteria for this study’s literature stems from the pertinent research focusing on the automation of structural design optimization. Digital tools include BIM for optimizing and automating building layout design and structural elements to minimize cost [54]. Consequently, the existing research documents are categorized based on structural design optimization and automation efforts targeting diverse project phases and process levels, including the conceptualization and configuration, design [52], automated code compliance [55,56], prefabrication, and automation, which raise accuracy, productivity, and reduce waste [57]. Digital tools are also used in the construction execution [58] and structural monitoring phase [59]. While there exist multiple stages in AEC projects where SDO efforts could streamline work processes, this study focuses on the systems and processes typically influenced by early stage SDO practices. Furthermore, SDO applications extend to various structural systems such as steel structures [60], composite structures, and pre-stressed forms; however, this study confines itself to reinforced concrete (RC) structures: the most prevalent structural form that may exhibit specific configurations during the design stage.

### 2.2. PRISMA Workflow

Similarly to David A. et al. [61], this study follows a step-by-step workflow to retrieve the literature pertaining to the subject matter, utilizing specific keywords to filter research documents in line with the inclusion and exclusion criteria, depicted in Figure 1.



The following steps outline the process of implementing the PRISMA methodology in this study.



**Figure 1.** Description of procedural steps involved in the PRISMA research approach for literature retrieval, filtering, inclusion and exclusion criteria, and categorization of research records. (Figure source: authors).

**Step (1) Literature Search Process:** The methodology for the literature search has a profound impact on the outcomes of systematic reviews [62], particularly in emerging research fields such as the automation of design processes in the AEC sector. This study adopts a rigorous approach to retrieve the literature from reputable scientific research databases, including Scopus, Web of Science (WoS), Springer, Taylor and Francis, and the ASCE Library. All these databases are chosen to cover a maximum amount of relevant research records and offer a comprehensive overview of State-of-the-Art research in automating structural design optimization. Notably, Google Scholar is excluded from the review due to its unsuitability for systematic reviews [63]. The search queries used during this step were as follows: (“BIM-based” OR “automated” OR “BIM-assisted” OR “advanced” OR “intelligent” OR “integrated”) AND (“reinforced concrete structural design” OR “RC design” OR “structural systems” OR “structural patterns” OR “multi-objective”) AND (“optimization” OR “optimisation” OR “optimum” OR “optimal”) AND (“framework” OR “approach” OR

“technique” OR “algorithms” OR “methods” OR “procedures”). These queries aim to identify articles published between 2010 and July 2023, encompassing a significant volume of the literature in this research domain. The initial emphasis is on the Scopus database, which yielded a total of 319 records, followed by other databases, which yielded an additional 293 articles.

**Step (2) Literature Inclusion and Exclusion Criteria:** This stage involves a comprehensive review of each article, from the abstract to the conclusion sections. A total of 612 documents are collected from all the databases mentioned above. The screening process is carried out in three phases, commencing with the elimination of duplicate findings and studies from other industries, resulting in 379 articles. Subsequent screenings evaluate the titles and abstracts of publications, leading to the exclusion of papers centered on structural systems beyond this study’s scope. Additional records not meeting the eligibility criteria outlined in this study’s scope and not aligning with the service or functional area for which the SDO study is intended for are excluded in the final phase. Ultimately, 218 records remain for a full evaluation.

**Step (3) Literature Categorization:** The final step involves categorizing the documents for a comprehensive full-text review. The available research records are categorized based on their primary contributions to structural design optimization and automation, targeting distinct project phases and process levels. These categories encompass the *conceptualization and configuration stage*, *automated code compliance stage*, *fabrication and prefabrication details*, *construction execution stage*, and *structural monitoring phase* (further elaborated upon in the Section 3.2). Table 1 succinctly summarizes the steps, including search strings, filtering, inclusion and exclusion, and categorization, involved in the SLR review of the available research on SDO.

**Table 1.** Literature search parameters and conditions involved in the SLR review process.

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Source Databases</b>        | Scopus, Web of Science (WoS), Springer, Taylor & Francis, and ASCE Library                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Search String</b>           | (TITLE (BIM-based) OR TITLE (automated) OR TITLE (BIM-assisted) OR TITLE (advanced) OR TITLE (intelligent) OR TITLE (integrated) AND TITLE (“reinforced concrete structural design”) OR TITLE (“RC structural design”) OR TITLE (“RC design”) OR TITLE (“structural systems”) OR TITLE (“structural patterns”) OR TITLE (multi-objective) AND TITLE (optimization) OR TITLE (optimization) OR TITLE (optimum) OR TITLE (optimal) AND TITLE (framework) OR TITLE (approach) OR TITLE (technique) OR TITLE (algorithms) OR TITLE (methods) OR TITLE (procedures)) |
| <b>Time Period Restriction</b> | 2010–2023                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Article Types</b>           | Journal, Conference Paper, Book Chapter, Review                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Language Restriction</b>    | English                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Included Subject Areas</b>  | Engineering, Computer Science, Energy, Mathematics, Environmental Science, Decision Sciences, Business, Management and Accounting, Materials Science, Economics, Econometrics and Finance, Multidisciplinary                                                                                                                                                                                                                                                                                                                                                    |
| <b>Excluded Subject Areas</b>  | Physics and Astronomy, Social Sciences, Earth and Planetary Sciences, Chemical Engineering, Chemistry, Agricultural and Biological Sciences, Neuroscience, Medicine, Biochemistry, Genetics and Molecular Biology                                                                                                                                                                                                                                                                                                                                               |
| <b>Work/Area Industry</b>      | AEC, Construction, Structural Engineering, Civil Engineering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

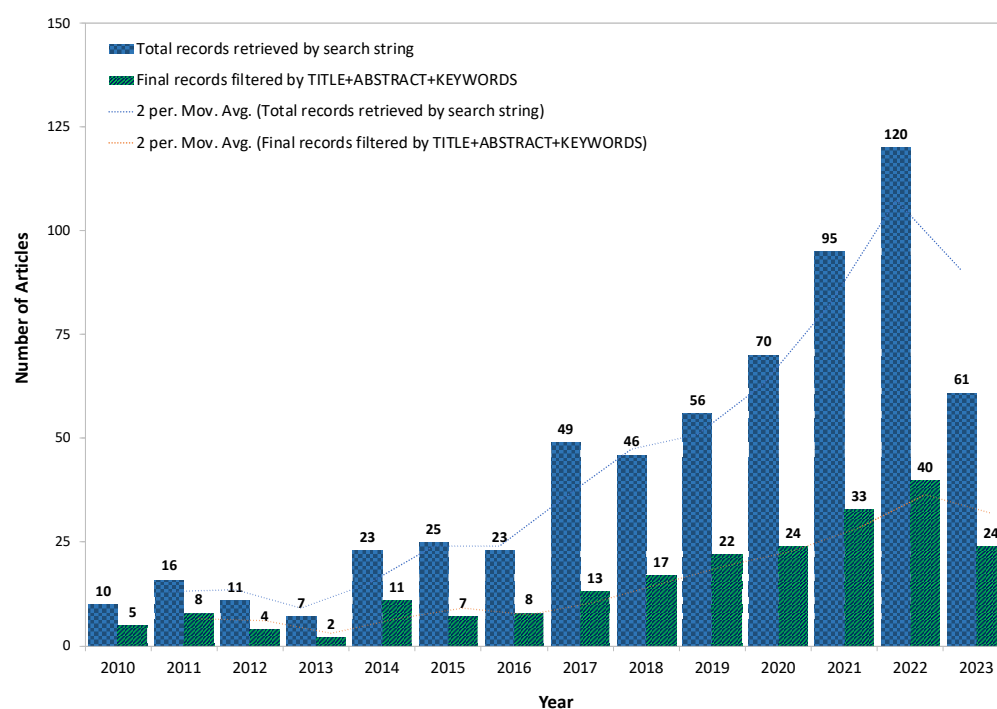
### 3. Results and Discussion

This section unveils the outcomes derived from the retrieved literature through both quantitative and qualitative analyses. Furthermore, it presents the results of an online

opinion survey targeting professionally accredited structural engineers in Italy and globally. These insights provide a practical foundation for comprehending the hurdles encountered while coordinating architects and structural engineers for SDO processes. The culmination of these analyses and survey inquiries has catalyzed the authors' inclination towards devising a structured framework to automate and enhance early stage structural design optimization processes, fostering improved collaboration between architects and structural engineers within the AEC sector.

### 3.1. Quantitative Analysis of Current State-of-the-Art Concepts

From 2010 to 2023, quantitative analysis exposes a substantial growth in the volume of publications pertaining to automating SDO using digital tools to enhance collaboration. Figure 2 illustrates the escalating trend of this research field, particularly in recent years, signifying its mounting popularity among global researchers. This surge in digitized SDO endeavors could be attributed to a convergence of factors, including technological advancements and evolving demands within the modern AEC industry. Notably, the proliferation of advanced AI and ML algorithms has reshaped how designers and engineers analyze intricate structural data, leading to more efficient and precise optimization techniques [26]. For example, Lou et al. [64] combined the optimization algorithm and support vector machine (SVM) to optimize the shear wall conceptual design using historical data and reduce the cost of the real model. The structural weight could be automatically minimized based on the story drift and period ratio.



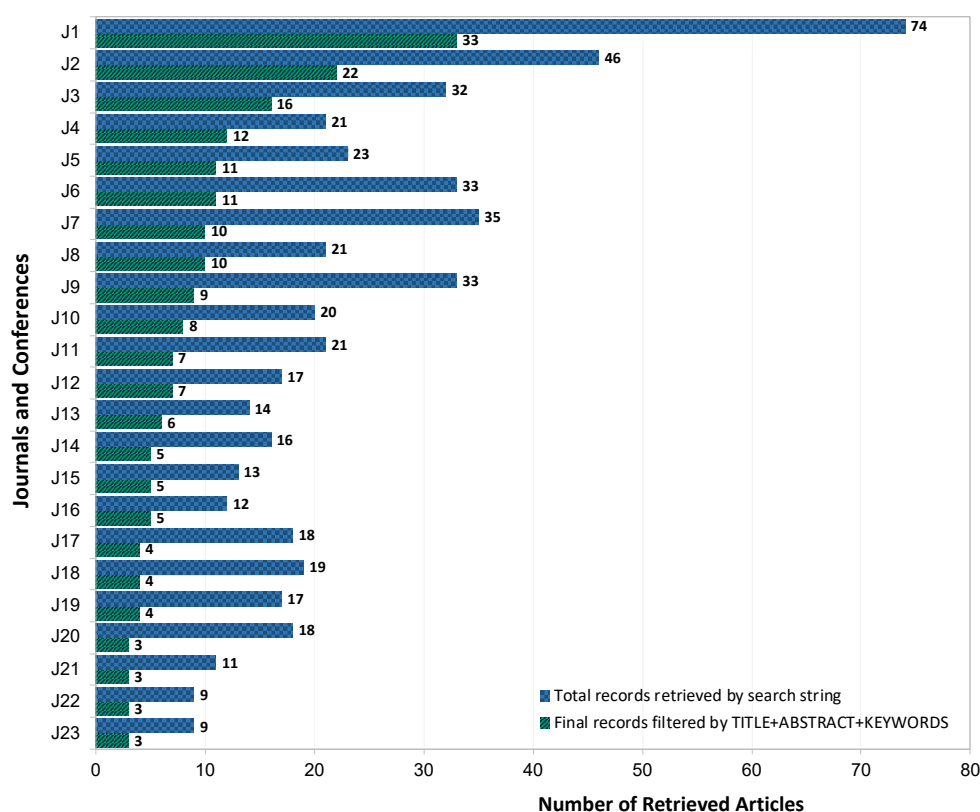
**Figure 2.** Number of articles on structural design optimization and automation per year between 2010–2023. (Figure source: authors).

The AEC sector's landscape is also witnessing a surge in innovative architectural projects, demanding cutting-edge optimization and automation to address complex geometries and load-bearing prerequisites. Traditional manual design methods struggle to match the precision and speed mandated by these visionary ventures. Furthermore, structural optimization and automation play a pivotal role in crafting environmentally sustainable structures with reduced carbon footprints, addressing the construction industry's growing sustainability concerns [65]. This push drives architects and engineers to explore sustainable materials, lightweight structures, and energy-efficient designs, aligning with the global



thrust towards eco-friendly construction practices. Additionally, heightened competition within the AEC sector fuels the impetus to optimize and automate key phases of projects, including structural designs. Construction firms embrace State-of-the-Art technologies to deliver superior projects faster and at lower costs than their competitors [3]. The surge in advanced SDO research in recent years finds its roots in technological progress, visionary projects, sustainability imperatives, and the pursuit of a competitive edge.

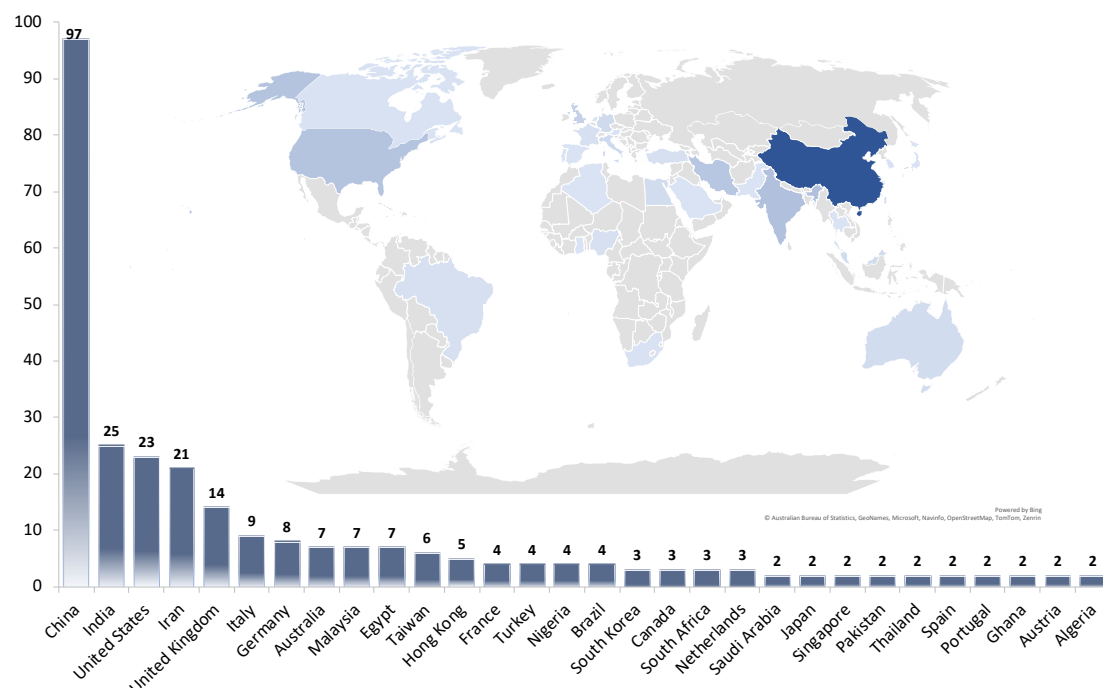
According to the search and selection criteria outlined in Table 1, several prominent journals have been identified as primary publishers of SDO research. The *Automation in Construction* journal published the most automating structural design optimization research (33 articles). This journal assumes a pivotal role in driving innovation across AEC practices and fostering the adoption of Construction 4.0 technologies by disseminating cutting-edge research spanning automation, robotics, AI, IoT, and other technological solutions. The *Journal of Cleaner Production*, renowned for their pioneering research in cleaner production methods, circular economy, renewable energy, and eco-friendly technologies across sectors, including construction, propels transformative change, houses 22 articles. Other notable contributors to SDO research include the *Journal of Building Engineering*, *Structural and Multidisciplinary Optimization*, and the *ISARC—International Symposium on Automation and Robotics in Construction*. Figure 3 encapsulates the major journals along with the corresponding number of articles, which are delineated by the search string and refined through selection and filtering criteria.



J1: *Automation in Construction*; J2: *Journal of Cleaner Production*; J3: *Journal of Building Engineering*; J4: *Energy*; J5: *MDPI Energies*; J6: *Engineering Structures*; J7: *Renewable and Sustainable Energy Reviews*; J8: *Procedia Engineering*; J9: *Building and Environment*; J10: *Construction and Building Materials*; J11: *Structural and Multidisciplinary Optimization*; J12: *Energy and Buildings*; J13: *MDPI Sustainability*; J14: *Procedia CIRP*; J15: *Structures*; J16: *ISARC—International Symposium On Automation And Robotics In Construction*; J17: *Computers & Structures*; J18: *Thin-Walled Structures*; J19: *Advances in Engineering Software*; J20: *Computer-Aided Design*; J21: *Advanced Engineering Informatics*; J22: *Expert Systems with Applications*; J23: *Engineering Optimization*

**Figure 3.** The number of collected and filtered articles in major journals and conferences. (Figure source: authors).

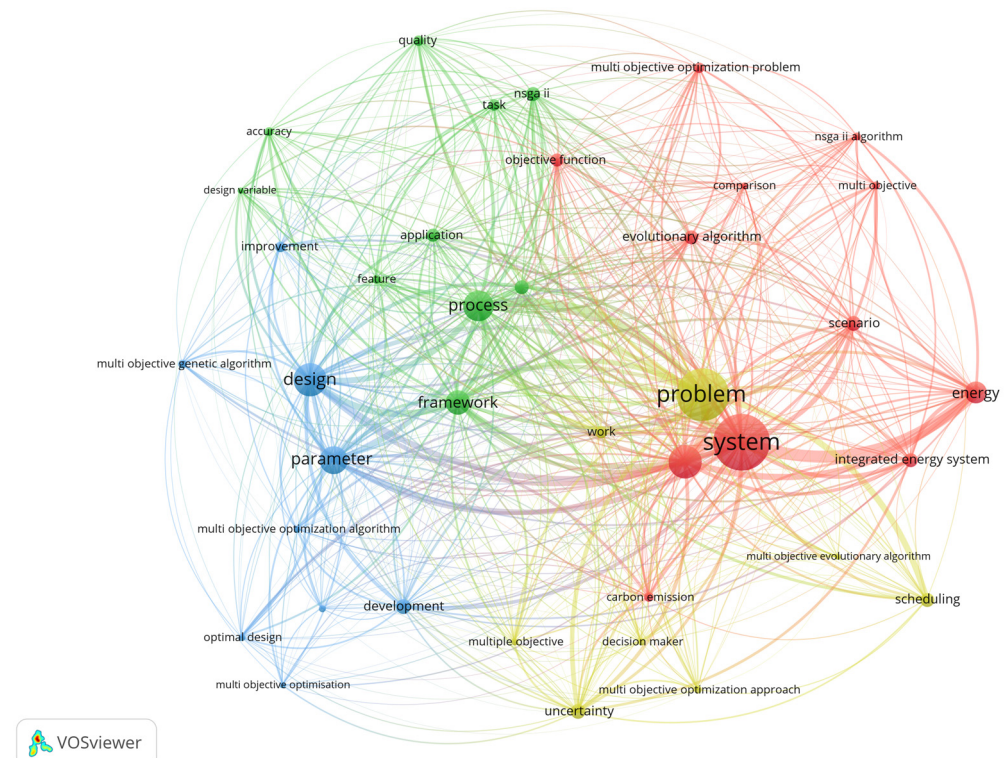
This study extends its rigorous analysis to the countries with the highest number of articles retrieved from 2010 to 2023. According to this analysis, *China* leads with 97 publication records, encompassing nearly 45% of the final documents. *India* takes second place, contributing 25 documents, a substantial portion of the total records (Figure 4). *China*, *India*, *the United States (US)*, *Iran*, and *the United Kingdom (UK)* collectively account for approximately 83% of the retrieved documents. Other countries, including *Italy*, *Germany*, *Australia*, and *Malaysia*, occupy the following spots in the list. The dominance of *China*, *India*, *the US*, *Iran*, and *the UK* is noteworthy, indicating their prominent roles in investigating the SDO topic. Chinese researchers' prevalence in the SDO domain can be attributed to various interlinked factors stemming from historical and contemporary contexts. Chinese researchers have committed to embracing technology across construction disciplines, recognizing technological advancement and innovation as key drivers of economic growth. For instance, a study by [19] harnessed BIM technology to design complex high-rise building envelope geometries, predicting structural behaviors and energy performances at the early design stage. Another Chinese study [21] proposed an AI-based intelligent methodology leveraging existing structural design datasets for shear walls, enabling the automatic design of optimal shear walls with an enhanced quality and performance in residential buildings. Notably, major funding sources for SDO research projects in China include the National Natural Science Foundation of China, the National Key Research and Development Program of China, the National Office for Philosophy and Social Sciences, the Natural Science Foundation of Shandong Province, and the China Scholarship Council.



**Figure 4.** Top countries that published structural design optimization research. (Figure source: authors).

The quantitative analysis extends to keyword occurrences, with VOSviewer 1.6.19 employed to consolidate keywords with more than 10 instances. In this context, 39 terms meet the criteria and are clustered into 4 groups in VOSviewer, as depicted in Figure 5. The clusters contain the following number of keyword items: *Cluster 1* (12 items): carbon emission, comparison, cost, energy, evolutionary algorithm, integrated energy system, multi-objective, multi-objective optimization, NSGA II algorithm, objective function, scenario, and system; *Cluster 2* (10 items): accuracy, application, decision, design variable, feature, framework, NSGA II, process, quality, and task; *Cluster 3* (9 items): design, development, improvement, integrated approach, multi-objective genetic algorithm, multi-objective optimization, optimal design, and parameter; *Cluster 4* (8 items): decision maker, multi-objective evolutionary

algorithm, multi-objective optimization approach, multiple objectives, problem, scheduling, uncertainty, and work. Keyword analysis highlights the recurring use of terms such as system, problem, process, and design, emphasizing the central role of automating SDO efforts in various stages and levels of AEC projects.



**Figure 5.** Keyword clustering analysis on the final articles. (Figure source: authors).

Furthermore, within the context of this research, which emphasizes SDO endeavors at project phases and systems, operations, and process levels, it is pivotal to dissect the motivations driving these efforts. During the era of Construction 4.0, architects and structural engineers are keen to adopt emerging digital tools and technologies that enhance systems, operations, and processes. This transformation redefines approaches to structural design optimization and automation [22]. For instance, practitioners recognize the potential of comprehensively analyzing building systems using digital tools and advanced algorithms, leading to more efficient and cost-effective designs for buildings and infrastructure [66]. Automating early stage structural design aspects unlocks the capability for process-level automation, replacing labor-intensive tasks with intelligent machines, mitigating human errors, and accelerating project delivery. An illustrative instance of process-level automation is outlined in a study by Liu Y. et al. [67], which utilizes BIM semantic information delivery to automate rebar fabrication processes, reducing waste and manual rebar bending time. Consequently, the enhancement of AEC projects at systems, operations, and process levels through the automation of early stage SDO engenders significant enhancements in the collaboration between architects and structural engineers, which addresses the integration challenges. Moreover, the keyword clustering during the analysis accentuates the preeminence of systems in the selected documents. Although keyword analysis offers a mapping of the reviewed literature, more nuanced exploration is required for emerging themes, such as automation, computational design, and generative design. Hence, in line with the articles' focus areas and contributions, a qualitative content analysis is undertaken to identify themes and patterns within the examined articles.

### 3.2. Qualitative Content Analysis

This qualitative analysis segment focuses on the early stage SDO efforts that facilitate the other lifecycle phases of the built asset across diverse systems, operations, and process levels. The gathered literature is segmented into the following thematic categories, which are aligned with the principal emphasis of the SDO approach, that target specific aspects of AEC projects:

**(C1) Conceptualization and Configuration Stage:** This category encompasses SDO efforts undertaken in the preliminary design and conceptualization phase. It involves evaluating various design parameters and investigating the influence of design variables and constraints. For instance, Liao et al. [21] proposed a generative adversarial network (GAN)—based shear wall design method that was learned from existing shear wall design documents. The structural design datasets were created by abstracting, classifying, and parameterizing them according to the height of the building and the seismic design category; then, these datasets performed structural design intelligently. The GAN model raised shear wall design proficiency using data and hyper-parametric analytics based on adversarial training. Aidy et al. [68] used genetic algorithms to optimize the concrete slab thickness of floors and of the steel bars of columns of a hospital based on labor costs, lateral earthquake load distribution, seismic loads, etc.

**(C2) Automated Code Compliance Stage:** In this category, SDO efforts are centered around automating code compliance checks during the design process. This involves verifying the adherence to codes related to safety, structural integrity, building regulations, and other AEC domain rules. Li et al. [69] proposed that exploratory genetic algorithms supported steel reinforcement rebar (part of reinforced concrete structures) design checking and optimization per building codes, thus automatically identifying optimal clash-free rebar design in compliance with code-stipulated requirements, while plummeting 75–90% of the computation time.

**(C3) Fabrication and Prefabrication Details:** This category pertains to the SDO studies focusing on providing comprehensive fabrication and prefabrication layout information. These efforts facilitate the smooth execution of construction processes. Baghdadi et al. [70] utilized particle swarm optimization (PSO), MATLAB, and the finite element software for optimizing beam layouts and their connections under free and irregular wall arrangements. The proposed technique optimized the structural performance and eased the fabrication process with minimum effort. Similarly, genetic algorithms optimized floor plans and construction, and revealed that the floor plan shape and window-to-wall ratio strongly correlated with a building's energy performance [71].

**(C4) Construction Execution Stage:** In this category, SDO initiatives are directed towards automating construction execution processes for structural systems in building assets or infrastructure facilities. Schwartz [72] presented a tool that automated the optimal building designs per lifecycle carbon footprint and lifecycle costs via a space-allocation generative-design application, NSGA-II optimization, and lifecycle performance per thermal simulations.

**(C5) Structural Monitoring/Operations Phase:** This category encompasses SDO endeavors that propose frameworks and approaches for monitoring the health and performance of structural facilities. This facilitates predictive maintenance strategies. Dam engineering, maintenance, and reinforcement (MAR) are essential for safeguarding project safety and for lengthening the service period. Yet, large-scale MARs of large-scale and long-term projects raise high costs. The whole lifecycle cost method that considers the design for the operation of a dam is introduced to lower costs and resources. The optimal initial design of the gravity dam and its optimal MAR plan are obtained through the optimization algorithm [73].

These thematic categories stem from the primary contributions of SDO efforts towards specific structural systems, operations, and process levels while aligning with the scope criteria defined within this study. For a more in-depth qualitative analysis and to propose a future effort of creating a BIM-based early stage sustainable structural design optimiza-



tion (ESSDO) framework, Table 2 presents further insights into the categorization, and is accompanied by the selected literature studies.

**Table 2.** SDO automation articles categorized per AEC project’s systems, operations, and processes.

| Category | Theme Description                                                                                                                                                                                    | Theme Sub-Categories                            | Research Contribution                                                                                                                                                                                                                                                            | Relevant Literature             |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| C1       | SDO efforts at the very early design and conceptualization stage, where the configuration of various design parameters is evaluated while inspecting the impact of design variables and constraints. | Construction Materials’ Optimization            | Research puts forth intelligent frameworks based on BIM to explore design detailing and configurations, and aims to optimize materials, cost, and time in structural design projects.                                                                                            | [4,6,9,16,17,23,68,69,72,74–95] |
|          |                                                                                                                                                                                                      | Construction Cost Optimization                  |                                                                                                                                                                                                                                                                                  |                                 |
|          |                                                                                                                                                                                                      | Energy (carbon emissions) Optimization          |                                                                                                                                                                                                                                                                                  |                                 |
|          |                                                                                                                                                                                                      | Planning and Scheduling Optimization            |                                                                                                                                                                                                                                                                                  |                                 |
| C2       | SDO efforts for streamlining the review of complicated rules and regulations.                                                                                                                        | Automated Code Compliance Checking              | Research studies’ BIM-centered automated code compliance checking methodologies, replacing traditional manual trial-and-error methods. These approaches enhance efficiency in verifying compliance with complex rules and regulations.                                           | [25,55,96–107]                  |
|          |                                                                                                                                                                                                      | Structural Design Authoring                     |                                                                                                                                                                                                                                                                                  |                                 |
|          |                                                                                                                                                                                                      | Automated Design Review and Evaluation          |                                                                                                                                                                                                                                                                                  |                                 |
|          |                                                                                                                                                                                                      | Rule-Based Generative Design Optimization       |                                                                                                                                                                                                                                                                                  |                                 |
| C3       | SDO studies provide information on fabrication and prefabrication layout information for executing the construction processes.                                                                       | Fabrication and Prefabrication                  | Research uses computer-aided design (CAD) methods to offer information about fabrication and prefabrication layouts. These methods enhance the precision of manufacturing and prefabrication processes for structural components.                                                | [27,67,74,108–116]              |
|          |                                                                                                                                                                                                      | Building Façade and Structural Components       |                                                                                                                                                                                                                                                                                  |                                 |
|          |                                                                                                                                                                                                      | Structural Design Layout                        |                                                                                                                                                                                                                                                                                  |                                 |
|          |                                                                                                                                                                                                      | Digital Manufacturing                           |                                                                                                                                                                                                                                                                                  |                                 |
| C4       | SDO studies automate the construction processes.                                                                                                                                                     | Automated and Robotic Construction              | Studies automate construction through BIM and advanced digital technologies, reducing time and minimizing waste.                                                                                                                                                                 | [10,33,117–130]                 |
|          |                                                                                                                                                                                                      | DfMA (Design for Manufacturing and Assembly)    |                                                                                                                                                                                                                                                                                  |                                 |
|          |                                                                                                                                                                                                      | Waste Management and Scheduling                 |                                                                                                                                                                                                                                                                                  |                                 |
|          |                                                                                                                                                                                                      | Construction Modelling (BIM 4D and 5D)          |                                                                                                                                                                                                                                                                                  |                                 |
| C5       | SDO studies for monitoring the building and infrastructure health and energy performance.                                                                                                            | Structural Health Monitoring                    | Scholarly works integrated BIM automation and other technologies for monitoring the health and performance monitoring of structural facilities. These might involve predictive maintenance, and contribute to the longevity and operational efficiency of the structural assets. | [5,12–14,24,131–143]            |
|          |                                                                                                                                                                                                      | Retrofitting Optimization                       |                                                                                                                                                                                                                                                                                  |                                 |
|          |                                                                                                                                                                                                      | Energy Performance                              |                                                                                                                                                                                                                                                                                  |                                 |
|          |                                                                                                                                                                                                      | Renovation Optimization                         |                                                                                                                                                                                                                                                                                  |                                 |
|          |                                                                                                                                                                                                      | Smart Infrastructure and Predictive Maintenance |                                                                                                                                                                                                                                                                                  |                                 |

Most articles (105) fall under the C1 category, which involves optimizing structural design configurations during the design stage, and account for almost half of the articles. This underscores the industry’s recognition of the importance of strategically addressing the challenges and design configurations in the early stages of a project, which offers manifold benefits throughout a building’s lifecycle [15]. Consequently, enriching early stage design decision making through SDO approaches contributes to safer structures, reduced maintenance costs, and adaptability to evolving needs, thereby ensuring sustained value for the stakeholders.

In contrast, the C2 category, related to automated code compliance checking, comprises only 14 documents, constituting less than 1% of the final documents. This observation serves as a call to action for the structural designers and engineers to explore this domain



and propose solutions for automatic code compliance. The integration of digitalization in SDO efforts, driven by complex modeling and design, introduces new challenges in the design collaboration and review [144]. Manual reviews of intricate norms and regulations have become error prone and time consuming, motivating the need for innovative solutions that automate code compliance. C3 (33), C4 (36), and C5 (30) follow in the number of articles retrieved pertaining to fabrication and prefabrication details, construction execution stages, and structural monitoring/operations phases, respectively.

The retrieved literature covers diverse areas of early stage decision making, including the optimization of construction materials, structural behavior, economic aspects, energy prediction and consumption, fabrication, and prefabrication of structural units, facade generation, and smart infrastructures through digital BIM and optimization methodologies. This research enhances the structural design assessment performance of building assets and infrastructure facilities, explores the design, and reduces the computational time for optimization, and consequently minimizes costly revisions. These SDO initiatives highlight that outputs during the early design stages significantly impact the entire lifecycle of building assets or infrastructure facilities. Such outcomes result in cost savings, enhanced efficiency, prolonged lifespan, and improved safety, all of which reverberate positively in the operational and maintenance phases of structural facilities. For instance, studies such as [9,10,69,81,119] propose BIM-based approaches to optimize steel reinforcement quantities in RC building structures while preserving allowable structural performance. These methodologies not only curtail material consumption but also automate SDO processes, significantly influencing the lifecycle of building structures. The literature search has resulted in retrieving almost half of the filtered studies targeting early stage decision making about structural design configurations. This might be because the optimized solutions at the earliest design stages of the design significantly impact the later stages up to the operations and the maintenance stages of a building asset or infrastructure facility. Moreover, Liu et al. [69,114] developed a BIM-based collaborative approach to accelerate the fabrication plan and data interoperability toward automatic prefabrication of optimized steel reinforcement. Similarly, the BIM-based parametric modeling [19,20,80,145] plays a substantial role in enhancing RC building designs during the early stages. Recognizing that measuring and managing embodied carbon from the outset of structural design projects unlocks emission reduction opportunities that are otherwise unattainable later.

Automation in the design stage using BIM and knowledge graphs transforms the construction industry, allowing automated code compliance checking [98], which is a pivotal driver of BIM-based automatic design review. In the structural engineering realm, reliance on 2D building plans for manual compliance checks with building regulations is time consuming and error prone [146]. While BIM models have been adopted for a design review in recent years, the complexity of models and diverse regional and international codes hinder the comprehensive research in this domain. The significance of SDO practices [75,101,111,117,123,125,126] in automating construction processes for complex structural systems, through 3D printing, prefabrication, additive manufacturing, and robotic construction, is acknowledged. However, the transition from research advancement to practical implementation is hampered by the complexity and information collaboration challenges.

Based on the literature review conducted above, it is evident that, despite notable progress in integrating various digital tools and methodologies within the domain of SDO, persistent challenges hinder the effective collaboration between architects and engineers during architectural modeling and structural design and analysis processes. This gap results in inefficient information exchange among stakeholders, leading to interoperability issues on both organizational and technical fronts. For instance, any alteration made to the architectural model necessitates a comprehensive rework, reanalysis, and rescheduling of the entire structural model when handed over to structural engineers. This underscores the urgent need for streamlined information transfer processes. While systematic reviews have offered valuable insights into the existing challenges associated with the coordination

between architects and structural engineers in SDO-based construction projects, they often fall short of delving into the real-world difficulties faced by these stakeholders. To bridge this gap between theoretical and practical perspectives within this research, it becomes imperative to conduct opinion surveys among industry experts and practitioners. Consequently, this study embarks on an opinion survey conducted among professional structural engineers and many other experts in the BIM and architecture disciplines in the European region to gain practical insights into the collaboration challenges encountered when employing digital tools, such as BIM, in structural design and analysis projects.

### 3.3. AEC Professionals Opinion Survey Results

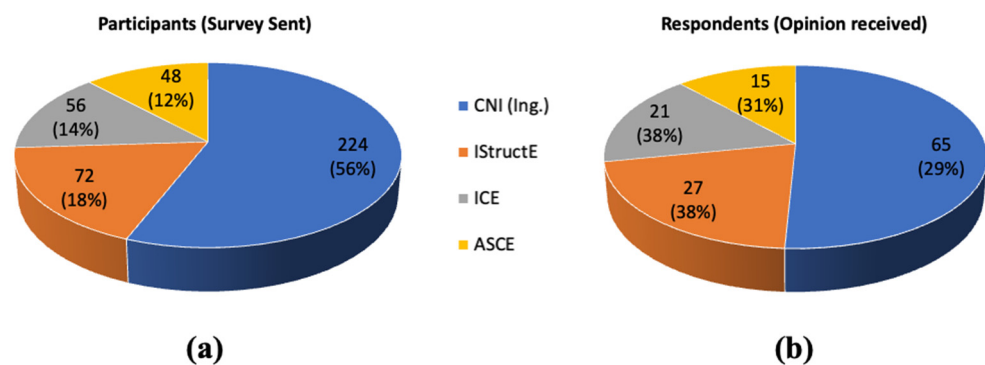
After conducting the PRISMA research, this study then conducts surveys with AEC professionals, especially in Italy and Europe. Surveys were commonly used to gauge industry opinion on various issues, such as AI [61], housing [147], and construction safety [148]. The objective is to explore their viewpoints on the challenges faced during the coordination between architects and structural engineers for structural design and analysis processes. The survey identifies challenges faced by these professionals in joint projects and gathers insights for solving these challenges. Online opinion surveys are a commonly used method to gather expert opinions, complementing the semi-structured expert interview approach [149]. The survey questionnaire focused on challenges in the structural design and analysis collaboration between architects and structural engineers, with an emphasis on interoperability issues and other impediments to their seamless teamwork. To broaden the participant pool, Italian engineers registered with local bodies, such as Consiglio Nazionale degli Ingegneri (CNI), Collegio degli Ingegneri (CdI), and Ordine degli Ingegneri (OdI), were initially invited. Additionally, structural engineers affiliated with renowned international organizations, such as the Institution of Structural Engineers (IStructE) in the UK, the Institution of Civil Engineers (ICE) in the UK, and the American Society of Civil Engineers (ASCE) in the USA, were contacted to contribute to the survey. Most of the participants registered with these global engineering governing bodies from Europe were invited not to limit the participants' pool only to Italy.

#### 3.3.1. Participants Profiles

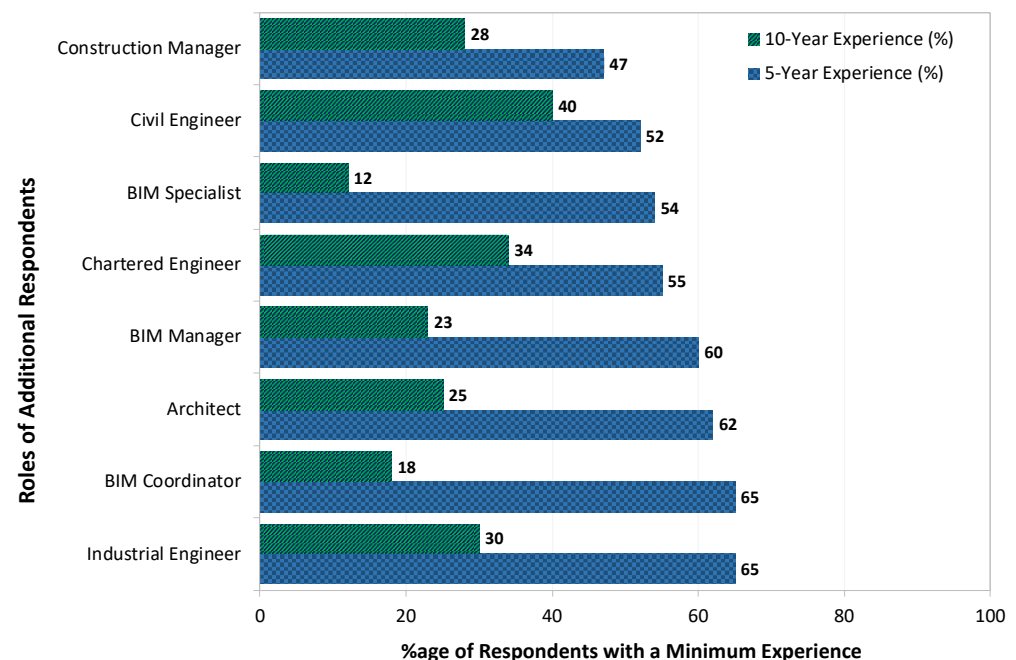
Various methods were employed to collect participants' contact details. Professional profiles and emails were gathered primarily through the LinkedIn search database. Additional information was sourced from publication records, regional chapters of engineering bodies, and networking connections. A total of 400 opinion surveys were distributed online, reaching engineers via email, LinkedIn connections, and personal connections. The distribution of contacted structural engineers and the response proportions are depicted in Figure 6. The participants' breakdown from the 400 surveys sent online is as follows (Figure 6a): 56% CNI-accredited engineers (Ing.), 18% IStructE professionals, 14% ICE engineers, and 12% ASCE-affiliated members. Of these, 128 responses were received, with the following distribution (Figure 6b): 29% CNI-accredited engineers (Ing.), 38% IStructE professionals, 38% ICE engineers, and 31% ASCE-affiliated members.

This study focused on spreading its population over Europe to increase the variety of participants and to have more global perspectives for meaningful research investigations involving regional engineering groups and companies across the continent. These entities possessed BIM expertise and hands-on experience in structural engineering tools such as Revit for Architecture and Structures, Autodesk RSA, and other relevant competencies. Snowball sampling was employed to gather additional data beyond the target group in Italy, extending to chartered engineers, construction managers, architects, civil engineers, and BIM roles (BIM manager, coordinator, and specialist). Participants received an introductory statement outlining the survey's goals, followed by a request for the information about their roles and experience. Consent for data processing was also obtained for this study's purposes. Figure 7 offers an overview of additional participant profiles approached through snowball sampling for data collection, including the percentage representation of

respondents with different levels of experience in various sectors of structural engineering (e.g., residential, commercial, high-rise buildings, industrial structures, and bridges).



**Figure 6.** Proportion distribution of (a) participants and (b) respondents in the opinion survey (Figure source: authors).



**Figure 7.** Additional participant profiles with corresponding years of professional experience (Figure source: authors).

### 3.3.2. Data Collection

To ensure the survey's reliability and relevance to the research objectives, a pilot study was conducted. Ph.D. students and academic instructors specializing in BIM and structural engineering from the Politecnico di Milano (Italy) and the University of Bologna (Italy) participated in the pilot opinion survey. Their feedback refined the survey questions, with a significant number of participants providing insights. The survey was administered in both English and Italian to accommodate native Italian respondents and English-proficient researchers from Europe. The survey questions served three purposes: (a) understanding collaboration challenges faced by architects and structural engineers during design and analysis, (b) exploring solutions for seamless integration between these professionals, and (c) gauging practitioners' perspectives on automated structural design and optimization using emerging methods.

### 3.3.3. Data Analysis and Results

A thematic analysis following the guidelines of Braun et al., [150] was employed to analyze the responses of the participants. The process involved familiarization of the data, identifying patterns, and extracting thematic observations related to the research questions. Key themes, such as inconvenient communication, data interoperability, design coordination, interdisciplinary communication, conventional collaboration, model complexity, problems in the integration of sustainability, resource constraints, etc., emerged from coherent response patterns.

An illuminating insight into the multifaceted panorama of challenges and prospective solutions within the collaborative realm of structural design and analysis was observed. Qualitative responses unearthed a spectrum of challenges and corresponding solutions within the realm of structural design, analysis, optimization, and interdisciplinary collaboration. These insights were organized into three distinct streams: generic, automated, and sustainable structural design and analysis. For a comprehensive insight into the thematic analysis findings, which intricately classify the identified issues and solutions and highlight the key aspects of structural design and analysis—including the generic, automated, and sustainable aspects of structural design and analysis—refer to Tables 3–5, correspondingly. These thematic categories impeccably aligned with this study’s core objective, which is to scrutinize challenges hindering collaboration and productivity within the AEC sector. Notably, prevalent challenges such as deficient communication, design coordination hurdles, and reliance on outdated structural engineering methods were acknowledged as commonplace issues. Their resolution necessitated the integration of advanced structural engineering design approaches and design optimization techniques.

**Table 3.** Thematic analysis of responses from the respondents about the issues faced during the integration between architects and structural engineers for structural design and analysis for *generic structural design and analysis*.

| Issues Theme                                | Reported Issues                                     | Solutions Theme                 | Suggested Solutions                                              | ESSDO Framework Contribution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------|-----------------------------------------------------|---------------------------------|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inconvenient Communication                  | Language and terminology differences                | Interdisciplinary Communication | Standardized nomenclature                                        | The ESSDO framework revolutionizes generic structural design optimization processes by streamlining communication and integration between architecture and structural engineering through centralization. It enhances design coordination and minimizes overdesign by early involvement of structural engineers, efficient error reduction, and advanced design optimization techniques. Additionally, it simplifies compliance with complex regional regulations, fostering environmentally conscious and cost-effective structural designs. |
|                                             | Lack of clear communication channels                |                                 | Properly centralized communication                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                             | Difficulty in sharing complex design configurations |                                 | Strategic organization of design details                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Design Coordination and Conflict Resolution | Excessive design changes and revisions              | Conceptual Design Coordination  | Involvement of SE in early design processes                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                             | Time-consuming coordination                         |                                 | Alignment of architectural and structural elements               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                             | Risk of design discrepancies                        |                                 | Reducing design errors early with efficient tools                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Overdesign                                  | Old methods of SE design and analysis               | Design Optimization             | Adoption of advanced methods of SE design                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                             | Manual calculation discrepancies                    |                                 | Utilization of design optimization                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Complex Regional Regulatory                 | Complex building codes                              | Regulatory Compliance           | Normalization through standard translation of complex regulatory |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                             | Challenges of frequent code changes                 |                                 | A generic code translation system to accommodate changes         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

**Table 4.** Thematic analysis of responses from the respondents about the issues faced during the integration between architects and structural engineers for structural design and analysis for *Automated Structural Design and Analysis*.

| Issues Theme                                                        | Reported Issues                                       | Solutions Theme                        | Suggested Solutions                                                           | ESSDO Framework Contribution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------------------------------------|-------------------------------------------------------|----------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Interoperability                                               | Different software platforms                          | Compatible Interoperability            | Acquiring software packages from the same provider company                    | ESSDO is an innovative and advanced framework approach to structural design optimization. It promotes compatible interoperability, customization, upgradation for legacy systems, and efficient adoption of automation through BIM-based software platforms for both architecture and structural engineering. The framework encourages iterative design, advanced visualization, and the use of optimization algorithms for accurate and optimized structural design outcomes. It sets itself apart by employing specialized BIM software, incorporating optimization algorithms, and spanning the entire lifecycle of building structures. ESSDO represents a transformative shift in structural design optimization and interactive integration between architects and structural designers during the construction projects at early phases. |
|                                                                     | Incompatible file formats and data structures         |                                        | Standardization of data schemas and formats                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                                     | Loss of information during data transfer              |                                        | Schemas with specialist in the SE                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Legacy Systems/Tools and Data Migration                             | Legacy design systems/tools and outdated data formats | Emerging Technologies                  | Customization of applications for given problems                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                                     | Data migration issues across software versions        |                                        | Smooth data migration to modern platforms                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                                     | Conversion issues for historical project data         |                                        | Solve interoperability                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Frequent Software Updates                                           | Difficulty in tracking design iterations              | Digital Design Iterations              | Validate and verify design versions                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                                     | Errors while working with outdated information tools  |                                        | Work with updated information tools                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Monotonous Design and Analysis                                      | Lack of design variations and options                 | Iterative Design and Analysis          | Use of parametric and generative modelling to explore multiple design options |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                                     | Possible conflicts on design options                  |                                        | Reduce conflicts through automated design iterations                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Conventional Collaboration                                          | Paper-based design data handling                      | Cloud-based Collaboration              | Use of cloud platforms to store, access, and share project data securely      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                                     | Time-consuming asynchronous collaboration             |                                        | Facilitate real-time collaboration across project players                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                                     | Local data handling issues                            |                                        | Eliminate the need for local data storage and minimize compatibility issues   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Two-Dimensional and Uncollaborative Three-Dimensional Visualization | 2D or non-interactive visualization platforms         | Advanced Visualization (Digital Twins) | Use advanced visualization tools to present design concepts                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                                     |                                                       |                                        | Leverage rapid prototyping and model creation                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                                     | Tedious modelling for built asset                     |                                        | Enhance project players' engagement through tangible representations          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                                     | Poor collaboration on digital models                  |                                        | Create digital replicas for real-time monitoring and analysis                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                                     |                                                       |                                        | Improve predictive maintenance and performance optimization                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |



Table 4. Cont.

| Issues Theme                  | Reported Issues                                               | Solutions Theme                             | Suggested Solutions                                                                                     | ESSDO Framework Contribution |
|-------------------------------|---------------------------------------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------------------|------------------------------|
| Legacy SE Design and Analysis | Slow and inaccurate design and analysis results               | AI and Optimization for Design and Analysis | Apply AI algorithms for faster and more accurate structural analysis                                    |                              |
|                               | Potential risks involved in design and analysis results       |                                             | Predictive modeling to identify potential design flaws early                                            |                              |
|                               | Imperfect manual safety and cost estimations                  |                                             | Optimize structural design for better safety and cost-effectiveness                                     |                              |
| Generic BIM Implementations   | Same BIM environment for architectural and structural designs | Rigorous BIM Capabilities                   | Employ specialized BIM software specifically tailored for structural design, analysis, and optimization |                              |
|                               | BIM for modelling purpose only                                |                                             | Implement optimization algorithms within the BIM environment                                            |                              |
|                               | Blind reliance on BIM design and analysis outcomes            |                                             | Move towards performance-based design approaches within BIM platforms                                   |                              |
|                               | BIM for SE design and analysis stages only                    |                                             | BIM for whole lifecycle of building structure                                                           |                              |

**Table 5.** Thematic analysis of responses from the respondents about the issues faced during the integration between architects and structural engineers for structural design and analysis for *Sustainable Structural Design and Analysis*.

| Issues Theme                                                          | Reported Issues                                                                                 | Solutions Theme                              | Suggested Solutions                                                                         | ESSDO Framework Contribution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Limited Awareness and Expertise in Sustainable SE Design and Analysis | Lack of awareness regarding sustainable design principles and practices                         | Raise Awareness and Sustainability Concerns  | Provide specialized awareness on sustainable design for architects and structural engineers | ESSDO transforms structural design optimization practices toward integrating sustainability concerns by raising awareness, promoting professional development, and integrating sustainability parameters associated with materials, building systems, and other aspects. It emphasizes sustainable material selection, advocates for streamlined sustainability concerns, and prioritizes structural resilience in the face of environmental challenges. ESSDO represents a pivotal shift in sustainable structural engineering design during the early stage design automation processes. |
|                                                                       | Insufficient expertise on sustainable building techniques                                       |                                              | Encourage professional development in green building practices                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                       | Difficulty in integrating sustainability into traditional design workflows                      |                                              | Foster collaboration with sustainability consultants                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Problems in Solo Integration of Sustainability in SE                  | Conflicting priorities between sustainable design and other concerns (cost, time, safety, etc.) | Balancing Sustainability with Other Concerns | Conduct lifecycle cost analysis that includes sustainable design                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                       | Challenges in convincing clients to invest in sustainable features                              |                                              | Showcase successful case studies of cost-effective sustainable projects                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                       | Identifying sustainable options within budgetary limitations                                    |                                              | Advocate for financial incentives to encourage sustainable SE designs                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

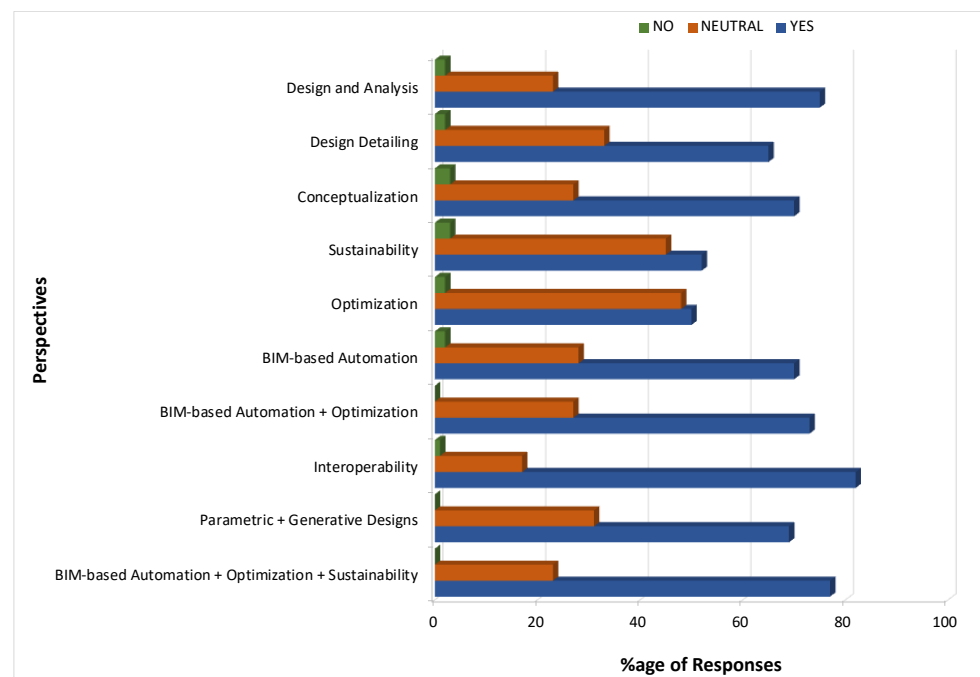
Table 5. Cont.

| Issues Theme                               | Reported Issues                                                                 | Solutions Theme                           | Suggested Solutions                                                                   | ESSDO Framework Contribution |
|--------------------------------------------|---------------------------------------------------------------------------------|-------------------------------------------|---------------------------------------------------------------------------------------|------------------------------|
| Conventional Sustainable Design Strategies | Lack of incorporating energy sources in building SE design                      | Integration of Renewable Energy Solutions | Use parametric design tools to optimize orientation of building structures            |                              |
|                                            | Infeasible structural elements with renewable energy systems                    |                                           | Integrate efficient renewable energy systems                                          |                              |
|                                            | Technical challenges of such integrations                                       |                                           | Merge renewable energy with building structural components                            |                              |
| Use of Excessive Steel and Concrete        | Lack of recommendation of sustainable materials with low environmental impact   | Material Selection and Lifecycle Analysis | Utilize databases and tools to assess environmental impacts of construction materials |                              |
|                                            | Lack of implementation of advanced principles                                   |                                           | Implement circular economy principles, promoting recycling and reusing materials      |                              |
|                                            | Incompatibility of sustainable materials with structural requirements           |                                           | Explore innovative and compatible sustainable materials                               |                              |
| Regulatory and Policy Constraints          | Complex and evolving sustainability regulations                                 | Sustainability Regulatory Compliance      | Advocate for streamlined green building regulation                                    |                              |
|                                            | Ignorance of design for safety and environment protection                       |                                           | Ensure design meet safety and environmental standards                                 |                              |
|                                            | Addressing inconsistencies in green building regulations                        |                                           | Establish partnerships with sustainability-focused organizations                      |                              |
|                                            | Compliance challenges with varying local, national, and international standards |                                           | Engage with policymakers and industry associations to influence sustainable policies  |                              |
| Vulnerability of Structural Systems        | Structural designs unable withstand climate change-related challenges           | Resilience and Climate Change Adaptation  | Employ climate modeling and risk analysis                                             |                              |
|                                            | Lack of long-term structural durability and adaptability                        |                                           | Adopt resilient design and construction techniques                                    |                              |

In the specific context of automated structural design and analysis, an apparent concern emerged—the seamless transfer of data between architects and structural engineers. Addressing this challenge necessitated the creation of tailored applications and the establishment of a smooth mechanism for data migration. Solutions proposed by participants encompassed diverse dimensions. To foster sustainable structural design, recommendations included the incorporation of lifecycle cost analysis, the presentation of successful case studies, and the introduction of financial incentives. In the domain of material selection, leveraging databases and tools for assessing environmental impacts, coupled with embracing circular economy principles, emerged as effective strategies for suggesting environmentally conscious materials in the early stages of the design process [88]. Moreover, the integration of renewable energy solutions and sustainable materials aligned with the structural prerequisites and was deemed crucial in propelling the adoption of eco-friendly designs.

Quantitative analysis results, vividly depicted in Figure 8, underscored the strong inclination to complement structural design and analysis with advanced methodologies

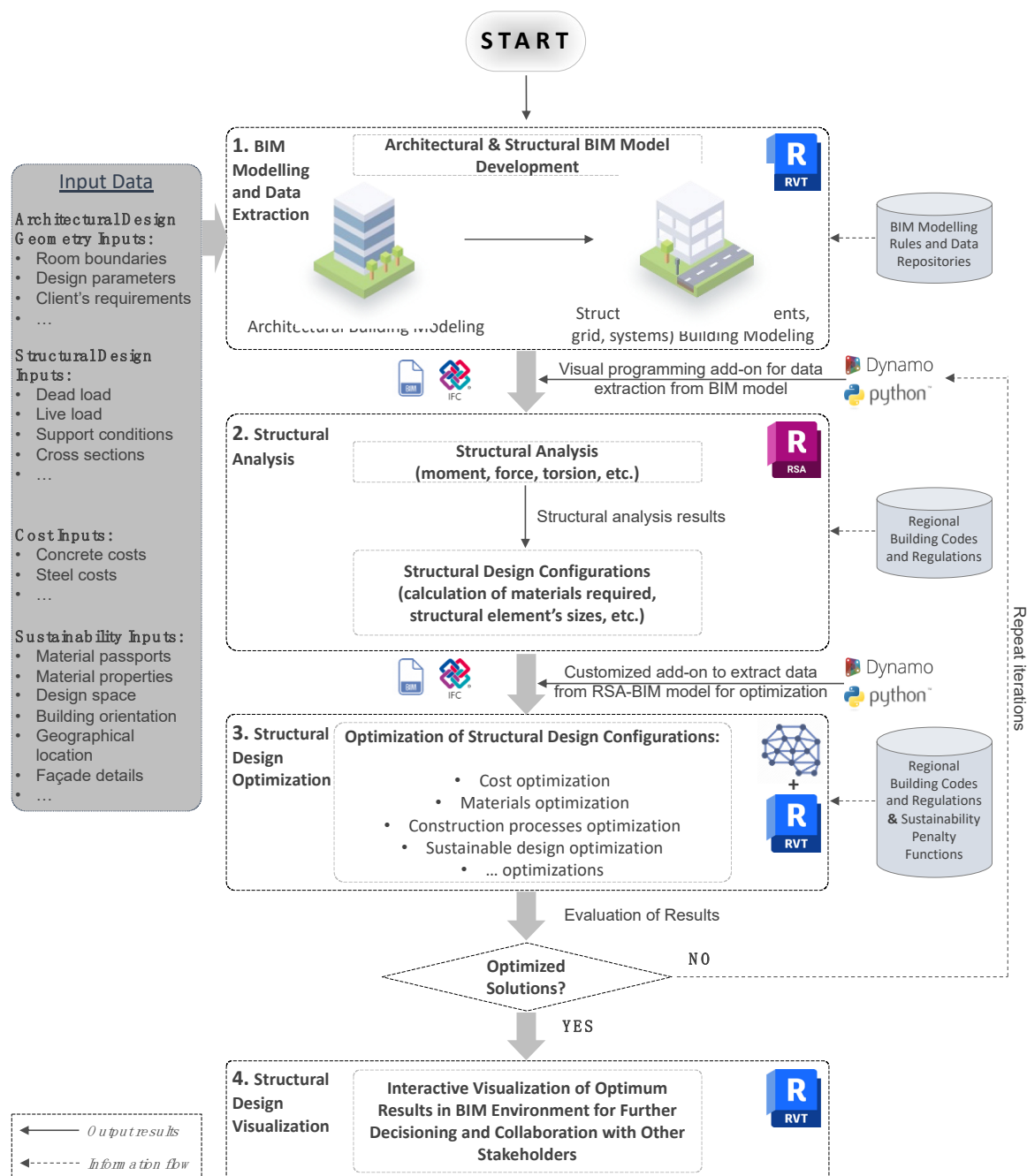
encompassing sustainability, automation, and interoperability. Remarkably, the survey participants unanimously favored steering clear of neglecting the amalgamation of these elements in the initial design phases. It is worth noting that the survey design did not encompass the in-depth investigation of obstacles and intricacies linked to implementing these advanced synergies. Interoperability has received a significant concern from the opinion perspectives as it is already the concern of the AEC sector when sharing digital information between project players, and many problems are usually discovered within, such as misinterpretation, data loss, and inaccuracy. For instance, BIM-based structural design automation complemented by sustainability integration in early design stages has also received a significant perspective of inclusion in the early stages of design for SDO.



**Figure 8.** Participants' perspectives on different advancements in SDO at early stages (Figure source: authors).

#### 4. Framework Proposal for BIM-Based Early Stage Sustainable Structural Design Optimization (ESSDO)

The main purpose of proposing a BIM-based framework for early stage sustainable structural design optimization (ESSDO) is to offer seamless integration between architecture and structural engineering works through automation to enhance design and analysis processes. ESSDO also provides the flexibility of integrating sustainability concerns at the early stages of structural design and analysis processes to obtain optimum solutions for design configurations that exhibit minimized environmental impacts by reducing construction material quantities while maintaining structural safety by following regional building codes. Although ESSDO could be slightly comprehensive to adopt, it offers customization of the processes according to the subject matter, which is an important aspect of the framework. The ESSDO framework shown in Figure 9 was developed and proposed to illustrate the viability of automating and streamlining the coordination between project players by connecting the structural and architectural models, allowing engineers to create structural designs based on the input data from the architectural model, such as geometry, design specifications, material properties, material costs, and structural elements. Then, ESSDO analyzes and optimizes generated models within architectural and structural BIM environments to deliver the results to structural designers/engineers for additional optimization and detailed designs.



**Figure 9.** Development and proposal of an intelligent framework for BIM-based early stage sustainable structural design optimization (ESSDO) to streamline the interactive collaboration between architects and structural engineers. (Figure source: authors).

The ESSDO framework comprises four pivotal components:

**BIM Modelling and Data Extraction:** Commencing with modeling architectural and structural designs for the targeted building structure or infrastructure facility, this phase generally falls within the domain of architectural works' project players. Architects, including structural engineers, typically resort to Autodesk Revit for Architecture and Structures, a widely employed BIM tool for architectural and structural designing applications. The architectural design environment is aptly facilitated by Revit. During BIM modeling, meticulous attention is given to imbuing the model with vital information encompassing architectural attributes, spatial organization, material costs, structural attributes, and material selection. Dynamo is harnessed to extract essential data from Revit into Dy-

namo, addressing the interoperability challenge via application programming interface (API) capabilities.

**Structural Analysis:** Parametric data extracted in the previous stage are channeled through BIM-supported file formats, such as Industry Foundation Classes (IFC) or Information Delivery Manual (IDM), to execute structural analysis and ascertain structural attributes within the RSA. RSA software equips structural engineers to undertake sophisticated BIM modeling and analysis for diverse building structures and infrastructure facilities. RSA fosters a more collaborative workflow and interoperability by interlinking bidirectionally with Autodesk tools such as Revit and Dynamo in three dimensions (3D). This step employs Dynamo's structural analysis package within RSA software, undertaking structural analysis and generating calculations for structural attributes. These calculations serve as a foundation for calculating requisite material quantities and sizes to withstand analyzed outcomes. By employing Python scripts within the Dynamo API, the computed results are conveyed back to Dynamo to fuel optimization across various objective functions, encompassing the maximization of building asset sustainability.

**Structural Design Optimization:** This phase hinges on insights garnered from structural analysis results. It entails determining requisite construction material quantities in alignment with regional building codes and standards. An array of options pertaining to material quantities and associated properties, structural geometries and performance, facade designs, structural element orientations, and structural systems are presented for selection within the design realm. Driven by the overarching design objective and defined objective function, structural design optimization takes flight, and is propelled by a suite of mathematical and optimization algorithms (SDO algorithms are extensively studied in [6]). Of notable significance, the integration of sustainability aspirations incorporates a pioneering penalty function designed to forestall overdesign and undue stress on structural elements. This pivotal step aspires to yield models that embody stability, safety, resilience, and cost effectiveness, thus echoing the essence of sustainable design practices. The optimization process unfurls iteratively until preset criteria are met, at which point the optimization ceases.

**Structural Design Visualization:** Upon the culmination of the optimization processes, the concluding phase of the framework involves updating structural design configurations within the BIM environment. This final visualization proves instrumental in facilitating interactive collaboration amongst stakeholders. It assumes a critical role in enriching decision-making processes, thereby augmenting the coordination between architects and structural engineers during the design and construction journey.

In summation, the ESSDO framework emerges as an innovative methodology for structural design that accords paramount importance to factors such as cost, time, and material efficiency. The integration of sustainability considerations across the early stage design process is emblematic of the framework's comprehensive scope. By harnessing the prowess of BIM technology and optimization strategies while earnestly addressing environmental impacts, ESSDO envisions a modern era characterized by sustainable and ecologically mindful building and infrastructure design. With the innate parametric attributes of architectural and structural models during the preliminary stages of ESSDO and an automated workflow that persists across the framework's various stages, any modifications to the Revit architectural model are automatically synchronized with Dynamo. This, in turn, generates novel optimized structural designs within RSA. This synthesis of BIM modeling and visualization platforms fortifies collaboration, streamlines workflows, and bolsters coordination between architects and structural engineers, culminating in improved design outcomes and enhanced efficiency throughout construction.

Hence, the ESSDO framework introduces a groundbreaking approach for early stage conceptualization, assessment, and optimization of RC structural designs. Furthermore, integrating architecture and structural engineering through automation enhances design and analysis processes while incorporating sustainability. This empowers engineers and designers to explore diverse design alternatives and select optimal solutions, thus leverag-



ing the potential of automation. The integration of automation and parametric modeling ensures real-time updates, dynamically generating optimized structural designs in response to changes. Focused on RC structures, this study provides a proof of concept, informed by a thorough literature review and practitioner input. The ESSDO enriches prevailing SDO practices by seamlessly integrating sustainability from the outset, fostering efficient, safe, and sustainable structures in alignment with global environmental goals. This BIM-based automation and optimization synthesis elevates collaboration, streamlines workflows, and enhances design outcomes and construction efficiency. Thus, ESSDO stands as a potent methodology and framework to address modern construction challenges while forging sustainable and impactful structures.

## 5. Research Gaps and Future Scope

The existing research in BIM-based sustainable SDO has made significant contributions to enhancing construction projects. The integration of sustainability considerations, innovative technologies, and optimization algorithms in SDO helps construction stakeholders. While the existing research has made significant progress, there are still challenges and limitations that have to be addressed. Challenges include fragmented collaboration between architects and structural engineers and interoperability problems between different software platforms.

Furthermore, there are limitations regarding the applicability of the existing SDO frameworks. Many studies have focused on specific aspects of SDO or limited the use of certain building typologies, which restricts generalizability. Moreover, the integration of sustainability considerations into SDO requires further exploration to ensure that the optimization process aligns with the environmental goals.

Based on the analysis of the existing research and the survey results, there are several research voids that can be explored. First, most research focuses on one specific building type. There is a need for a comprehensive framework that caters to different building types and accommodates various sustainability criteria.

Second, this article highlighted the challenges of communications between architects and structural engineers. The SDO frameworks should also facilitate communication and collaboration between different stakeholders, including architects, structural engineers and (possibly) AI—in view of the fast growth of generative AI tools.

Third, while different parts of the world are performing research on SDO, with some having performed more research than in the other parts of the world, no research investigated the impact of culture on BIM-based SDO usage, development, and adoption. As previous research showed that some cultures may accept innovation more [151], an evaluation of cultural impacts on SDO application and research may explore the impact of culture on BIM-based SDO development.

Furthermore, most research at present sheds light on one aspect of BIM-based SDO, and future research can also explore the integration of SDO frameworks with other domains, such as energy optimization, construction scheduling, and facility management, to achieve a more integrated and holistic approach to building design and operation.

## 6. Conclusions

This paper presents an advanced study focused on optimizing early stage structural design and analysis within BIM-based practices in the construction sector; this was performed by presenting this structural design optimization (SDO) realm from the viewpoint of the existing research review and the practical industry perspectives.

- An extensive literature review was conducted, which examines relevant research efforts and initiatives on the automation and optimization of structural design to establish a context for analyzing the integration of architecture and structural engineering fields. Employing both quantitative and qualitative methods, this analysis forms a base of knowledge for comprehending SDO practices and challenges faced by the architects and structural engineers during their coordination at the early phases of

design. This study brings attention to the research gaps, particularly in the overlooked domain of automated code compliance, prompting the need for further investigation.

- To address the challenges that architects and engineers encounter in automated SDO collaboration, an online survey is conducted among accredited structural engineers and BIM practitioners in order to collect practical perspectives from industry professionals. The survey's outcomes underscore interoperability as a crucial concern, echoing the challenges witnessed in the AEC sector concerning digital information exchange.
- This paper presents both quantitative and qualitative findings while also elucidating how the ESSDO framework tackles the challenges pinpointed in the survey. The proposed early stage sustainable structural design optimization (ESSDO) framework responds to the absence of automated and interactive collaboration in BIM-integrated SDO processes. The framework automates structural design, analysis, and optimization tasks by incorporating visual programming within a widely used BIM platform. It effectively addresses the interactive collaboration obstacles between architects and engineers.
- Additionally, integrating sustainability principles augments SDO by seamlessly embedding these principles from the outset, fostering the development of efficient, secure, and sustainable structures. The ESSDO framework synchronizes the parametric data between architectural and structural models, facilitating dynamic collaboration between architects and engineers. Any changes made to the architectural model are promptly reflected in the corresponding structural models.
- The core objective of this framework is to streamline and automate the structural design optimization process. It is designed to be user friendly, catering to individuals regardless of their programming expertise. The framework's scope is currently delimited to reinforced concrete (RC) structures. Nonetheless, ongoing efforts are dedicated to expanding its applicability for validation in real-world building structures or infrastructure facilities.

While this study introduced an advanced ESSDO framework to enhance the coordination between architects and structural engineers and to facilitate interactive integration, certain limitations should still be acknowledged. Numerous coordination issues were identified through an online survey, and industry professionals proposed various solutions. Nevertheless, it was not necessary to incorporate all these solutions within the ESSDO framework. For instance, industry experts suggested integrating sustainability regulations into SDO practices to promote sustainability. However, these regulations fall outside the scope of the ESSDO framework and should be subject to external regulatory oversight by organizations or countries. Furthermore, raising sustainability awareness among professionals during SDO, although it is a valuable concept, represents a theoretical rather than practical or technical solution. Consequently, it does not form part of the ESSDO framework. However, ESSDO offers flexibility by allowing practitioners to input various sustainability perspectives according to the specific nature of the problem and the configuration of building systems in their use of the framework. It is important to recognize that the extent to which these solutions are incorporated into the ESSDO framework may depend on the specific requirements and challenges presented by the building systems under consideration. Anticipated future research and industry applications are poised to address these limitations.

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